

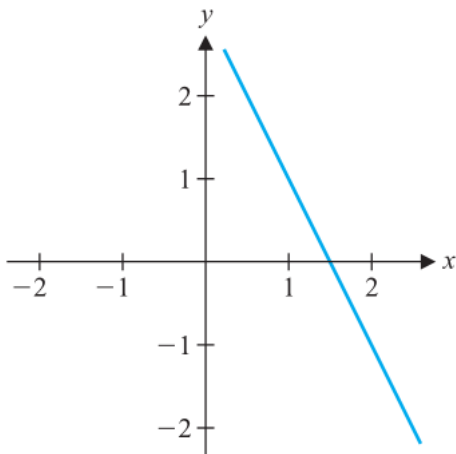
Lagrange Multipliers

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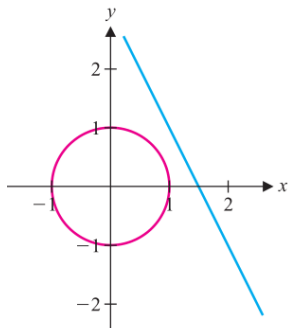
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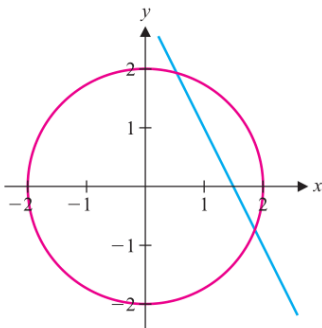
Find the point on the line $y = 3 - 2x$ that is closest to the origin.



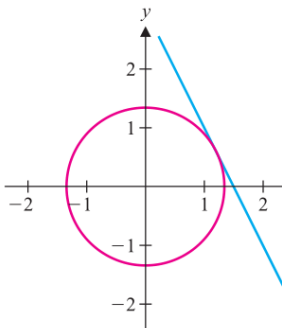
Closest point on the line $y = 3 - 2x$ from the origin.



$y = 3 - 2x$ and the circle of radius 1 centered at $(0, 0)$



$y = 3 - 2x$ and the circle of radius 2 centered at $(0, 0)$



$y = 3 - 2x$ and a circle tangent to the line

The closest point on the line $y = 3 - 2x$ from the origin will be greater than 1 and less than 2 units

Translating the preceding geometric argument into the language of calculus

- 1 Minimize the distance from the point (x, y) to the origin, given by $\sqrt{x^2 + y^2}$
 - $\sqrt{(x - x_0)^2 + (y - y_0)^2}$ with $(x_0, y_0) = (0, 0)$ gives $\sqrt{x^2 + y^2}$
 - Both the distance and its square are minimized at the same point always
- 2 Minimize $f(x, y) = x^2 + y^2$, subject to the constraint that the point lie on the given line

Minimize

$$f(x, y) = x^2 + y^2$$

subject to

$$g(x, y) = 2x + y - 3 = 0.$$

Generally

- $f(x, y)$ is called *objective function*
- $g(x, y)$ is called *constraint*

The Method of Lagrange Multipliers

Theorem (Lagrange Multipliers in $\mathbb{R}^2$)

Suppose that the **maximum/minimum** value of $f(x, y)$ subject to $g(x, y) = 0$ occurs at (x_0, y_0) . Then, there exists a constant λ (called a **Lagrange multiplier**) such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0).$$

In component form

$$\begin{aligned} \frac{\partial f}{\partial x} &= \lambda \frac{\partial g}{\partial x} & \text{or } f_x &= \lambda g_x \\ \frac{\partial f}{\partial y} &= \lambda \frac{\partial g}{\partial y} & \text{or } f_y &= \lambda g_y \end{aligned}$$

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

$$\left. \begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 0 \\ \frac{\partial \mathcal{L}}{\partial y} &= 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 0 \end{aligned} \right\} \begin{aligned} f_x &= \lambda g_x \\ f_y &= \lambda g_y \\ g(x, y) &= 0 \end{aligned}$$

Minimize

$$f(x, y) = x^2 + y^2 \quad \text{subject to} \quad g(x, y) = y + 2x - 3 = 0.$$

Lagrange System

$$\text{Set } \nabla f = \lambda \nabla g, [2x, 2y] = \lambda[2, 1]$$

$$\implies [2x, 2y] = [2\lambda, \lambda]$$

In component form

$$\left. \begin{array}{l} 2x = 2\lambda, \\ 2y = \lambda \end{array} \right\} \implies x = \lambda, y = \frac{\lambda}{2}.$$

Substitute into $g(x, y) = 0$:

$$\frac{\lambda}{2} + 2\lambda - 3 = 0 \implies 5\lambda = 6$$

$$\implies \lambda = 6/5.$$

Result

$$x = \lambda \text{ \& \; } y = \frac{\lambda}{2} \implies \boxed{\left(\frac{6}{5}, \frac{3}{5}\right)}$$

closest point on $y = 3 - 2x$ from origin.

Minimum distance:

$$D_{\min} = \sqrt{\left(\frac{6}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \frac{3\sqrt{5}}{5}.$$

Find the greatest and smallest values of

$$I(x, y) = xy$$

subject to $\frac{x^2}{8} + \frac{y^2}{2} = 1$.

Method: Lagrange Multipliers

Set up $\nabla I = \lambda \nabla g$, with

$$[y, x] = \lambda \left[\frac{x}{4}, y \right]$$

$$y = \lambda \frac{x}{4}, \quad x = \lambda y$$

Eliminate λ

$$\implies x^2 = 4y^2 \implies x = \pm 2y.$$

Using $x = 2y$ in $\frac{x^2}{8} + \frac{y^2}{2} = 1$

$$\frac{4y^2}{8} + \frac{y^2}{2} = 1 \implies y^2 = 1$$

$$\implies y = \pm 1, \quad x = \pm 2,$$

Using $x = -2y$ in $\frac{x^2}{8} + \frac{y^2}{2} = 1$

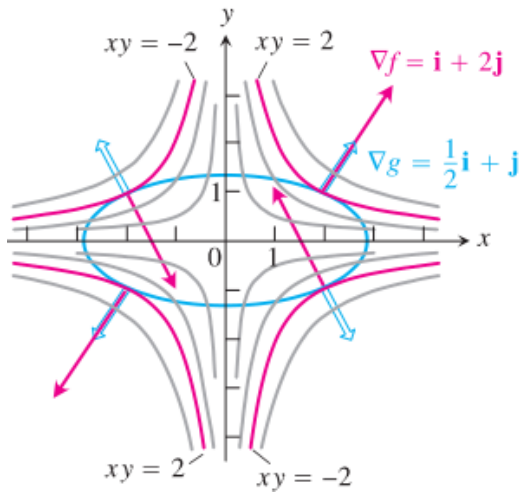
$$\frac{4y^2}{8} + \frac{y^2}{2} = 1 \implies y^2 = 1$$

$$\implies y = \pm 1, \quad x = \mp 2,$$

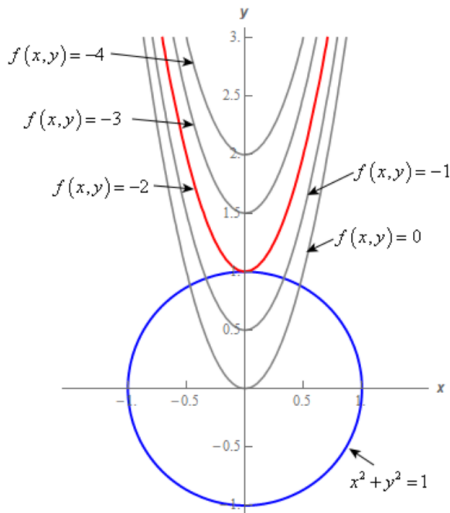
Maximum and Minimum Values

$$I_{\max} = 2 \quad \text{at } (2, 1), (-2, -1)$$

$$I_{\min} = -2 \quad \text{at } (2, -1), (-2, 1)$$



Find the minimum and maximum values of $f(x, y) = 8x^2 - 2y$ subject to the constraint $x^2 + y^2 = 1$.



Find the extremum of $f(x, y) = 8x^2 - 2y$ subject to $x^2 + y^2 = 1$.

Step 1: Lagrange Function:

$$\begin{aligned}\mathcal{L}(x, y, \lambda) &= f(x, y) + \lambda g(x, y) \\ &= 8x^2 - 2y + \lambda(x^2 + y^2 - 1).\end{aligned}$$

Step 2: System of Equations:

We calculate the partial derivatives:

$$\frac{\partial \mathcal{L}}{\partial x} = 16x + 2\lambda x = 0 = x(16 + 2\lambda)$$

$$\frac{\partial \mathcal{L}}{\partial y} = -2 + 2\lambda y = 0 \Rightarrow \lambda y = 1$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x^2 + y^2 - 1 = 0.$$

Step 3: Solve the System:

From $x(16 + 2\lambda) = 0$, we have two cases: $x = 0$ & $\lambda = -8$

Using $x = 0$, into $x^2 + y^2 = 1$,

we get: $y^2 = 1 \Rightarrow y = \pm 1$.

Substituting into $f(x, y)$:

$$f(0, 1) = -2, \quad f(0, -1) = 2.$$

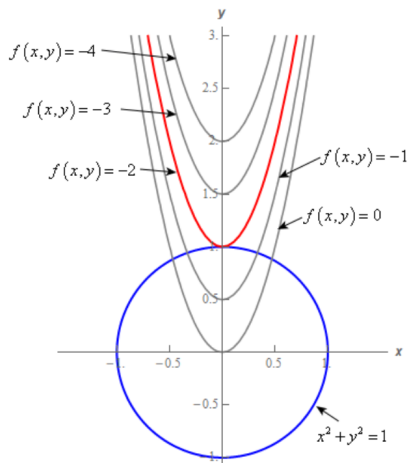
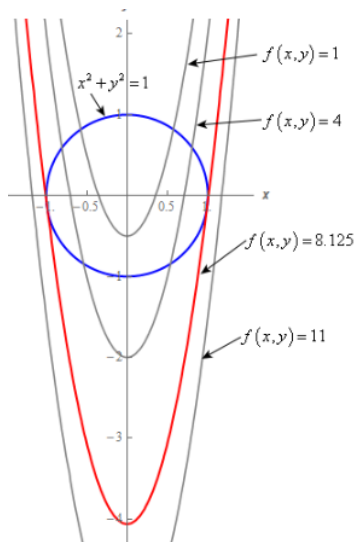
$$\lambda = -8 \text{ \& } \lambda y = 1 \implies y = -\frac{1}{8}$$

Using in $x^2 + y^2 = 1$, $\Rightarrow x = \pm \frac{\sqrt{63}}{8}$

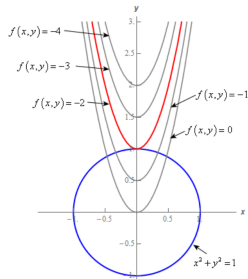
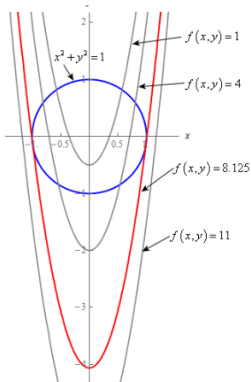
Step 4: Compare Values at CP :

$$f(0, \pm 1) = \mp 2, \quad f\left(\frac{\sqrt{63}}{8}, -\frac{1}{8}\right) = \frac{65}{8}$$

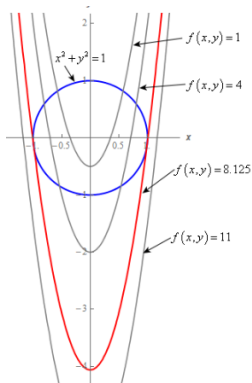
$$-2 \leq f(x, y) \leq \frac{65}{8}$$



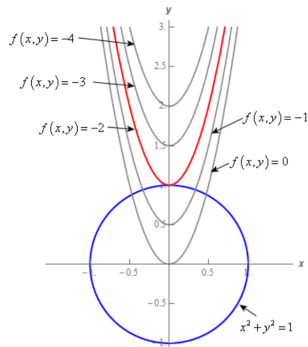
Sketch of the constraint as well as $f(x,y)$ for various values of k



The location of the minimum/maximum value of $f(x, y)$, must occur where the graph of $f(x, y)$ intersects the graph of the constraint when k is either the minimum or maximum value of $f(x, y)$.



- The graph of $f(x, y) = 8.125$ will just touch the graph of the constraint at two points
- The constraint are tangent at point of intersection



- The graph of $f(x, y) = -2$, that is, minimum value of $f(x, y)$ and point $(0, 1)$
- The two graphs at that point are tangent at point of intersection

Find the extremum of $f(x, y) = 3x + 4y$ subject to $x^2 + y^2 = 1$.

Step 1: Lagrange Function:

$$\begin{aligned}\mathcal{L}(x, y, \lambda) &= f(x, y) + \lambda g(x, y) \\ &= 3x + 4y + \lambda(x^2 + y^2 - 1).\end{aligned}$$

Step 2: System of Equations:

We calculate the partial derivatives:

$$\frac{\partial \mathcal{L}}{\partial x} = 3 + 2\lambda x = 0 = \frac{\partial \mathcal{L}}{\partial y} = 4 + 2\lambda y$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x^2 + y^2 - 1 = 0.$$

Step 3: Solve the System:

From $3 + 2\lambda x = 0$ and $4 + 2\lambda y = 0$,
we get: $\lambda = -\frac{3}{2x} = -\frac{2}{y} \Rightarrow 3y = 4x \Rightarrow y = \frac{4}{3}x$

Substitute into constraint:

$$x^2 + \left(\frac{4}{3}x\right)^2 = 1 \Rightarrow x^2 + \frac{16}{9}x^2 = 1$$

$$\frac{25}{9}x^2 = 1 \Rightarrow x^2 = \frac{9}{25} \Rightarrow x = \pm \frac{3}{5}$$

$$y = \frac{4}{3}x = \pm \frac{4}{5}$$

Evaluate $f(x, y)$:

$$f\left(\frac{3}{5}, \frac{4}{5}\right) = 3 \cdot \frac{3}{5} + 4 \cdot \frac{4}{5} = \frac{9}{5} + \frac{16}{5} = 5$$

$$f\left(-\frac{3}{5}, -\frac{4}{5}\right) = -\frac{9}{5} - \frac{16}{5} = -5$$

Step 4: Final Result:

Maximum value = 5, Minimum value = -5
Lagrange Multipliers

Lagrange Multiplier

Three Variables Case

If the two graphs are tangent at that point then their normal vectors must be parallel,

the two normal vectors must be scalar multiples of each other. Mathematically, this means,

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$$

$$\begin{aligned}\frac{\partial f}{\partial x} &= \lambda \frac{\partial g}{\partial x}, \\ \frac{\partial f}{\partial y} &= \lambda \frac{\partial g}{\partial y}, \\ \frac{\partial f}{\partial z} &= \lambda \frac{\partial g}{\partial z}.\end{aligned}$$

Find the maximum and minimum values of $f(x, y) = 5x - 3y$ subject to the constraint $x^2 + y^2 = 136$.

Lagrange multipliers

$$\nabla f = \lambda \nabla g$$

Gradients:

$$\nabla f = (5, -3), \quad \nabla g = (2x, 2y).$$

Equating gradients:

$$(5, -3) = \lambda(2x, 2y).$$

System of equations:

This gives the

$$\begin{cases} 5 = \lambda \cdot 2x & (1) \\ -3 = \lambda \cdot 2y & (2) \\ x^2 + y^2 = 136 & (3). \end{cases}$$

Solution Steps

From (1) and (2): $\lambda = \frac{5}{2x}$ and $\lambda = \frac{-3}{2y}$.

$$\frac{5}{2x} = \frac{-3}{2y} \implies 5y = -3x \implies y = -\frac{3}{5}x.$$

Substitute $y = -\frac{3}{5}x$ **into (3):** $\implies x^2 + \frac{9}{25}x^2 = 136 \implies \frac{34}{25}x^2 = 136.$

Solve for x^2 : $x^2 = 136 \cdot \frac{25}{34} = 100 \implies x = \pm 10.$

$$y = -\frac{3}{5}x \implies y = -6 \text{ when } x = 10, \quad y = 6 \text{ when } x = -10.$$

Evaluate $f(x, y)$:

$$f(10, -6) = 5(10) - 3(-6) = 68, \quad f(-10, 6) = 5(-10) - 3(6) = -68.$$

Conclusion: Maximum = 68, Minimum = -68.

Find the maximum and minimum values of $f(x, y) = 4x^2 + 10y^2$ on the disk $x^2 + y^2 \leq 4$.

Lagrange multipliers

$$\nabla f = \lambda \nabla g$$

Gradients:

$$\nabla f = (8x, 20y), \quad \nabla g = (2x, 2y).$$

Equating gradients:

$$(8x, 20y) = \lambda(2x, 2y).$$

System of equations:

This gives the

$$\begin{cases} 8x = \lambda \cdot 2x & (1) \\ 20y = \lambda \cdot 2y & (2) \\ x^2 + y^2 = 4 & (3). \end{cases}$$

Step 1: Critical Points

Find critical points within the disk.

$$f_x = 8x \implies 8x = 0 \implies x = 0.$$

$$f_y = 20y \implies 20y = 0 \implies y = 0.$$

Critical point: $(0, 0)$. This satisfies the inequality $x^2 + y^2 \leq 4$.

Solving for Critical Points

From $8x = 2\lambda x$:

$$2x(4 - \lambda) = 0 \implies x = 0 \text{ or } \lambda = 4.$$

Case 1: If $x = 0$, substitute into $x^2 + y^2 = 4$:

$$y = \pm 2.$$

From $20y = 2\lambda y$:

$$\lambda = 4 \implies y = 0.$$

Substitute into $x^2 + y^2 = 4$:

$$x = \pm 2.$$

Critical Points to Check:

$$(0, \pm 2), (\pm 2, 0), (0, 0).$$

Finding Maximum and Minimum Values

Evaluate $f(x, y)$ at the critical points:

$$f(0, 0) = 4(0)^2 + 10(0)^2 = 0 \quad (\text{Minimum})$$

$$f(2, 0) = f(-2, 0) = 4(2)^2 + 10(0)^2 = 16$$

$$f(0, 2) = f(0, -2) = 4(0)^2 + 10(2)^2 = 40 \quad (\text{Maximum}).$$

Conclusion:

- **Maximum Value:** 40 at $(0, \pm 2)$.
- **Minimum Value:** 0 at $(0, 0)$.

Optimization Problem Using Lagrange Multipliers

Optimize the function $f(x, y, z) = 4y - 2z$, subject to the constraints:

$$g(x, y, z) = 2x - y - z - 2 = 0 \quad \text{and} \quad h(x, y, z) = x^2 + y^2 - 1 = 0$$

Using Lagrange multipliers, we need to solve the following system:

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z) \quad \text{Two Lagrange multipliers } \lambda \text{ \& } \mu$$

$$g(x, y, z) = 0$$

$$h(x, y, z) = 0$$

Note that

The first equation really is three equations

Breaking Down the Lagrange Multiplier System

$$f(x, y, z) = 4y - 2z, \quad g(x, y, z) = 2x - y - z - 2 = 0, \quad h(x, y, z) = x^2 + y^2 - 1 = 0$$

$$\nabla f(x, y, z) = (0, 4, -2)$$

$$\nabla g(x, y, z) = (2, -1, -1)$$

$$\nabla h(x, y, z) = (2x, 2y, 0)$$

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z)$$

This becomes a system of equations

$$(0, 4, -2) = \lambda(2, -1, -1) + \mu(2x, 2y, 0)$$

Along with the constraints:

$$2x - y - z = 2$$

$$x^2 + y^2 = 1$$

Solving the System of Equations

$$\nabla f = \lambda \nabla g + \mu \nabla h$$

This becomes a system of equations

$$(0, 4, -2) = \lambda(2, -1, -1) + \mu(2x, 2y, 0)$$

Along with the constraints:

$$2x - y - z = 2$$

$$x^2 + y^2 = 1$$

We are given the system of equations:

$$0 = 2\lambda + 2\mu x \quad (1)$$

$$4 = -\lambda + 2\mu y \quad (2)$$

$$-2 = -\lambda \quad (3)$$

$$2x - y - z = 2 \quad (4)$$

$$x^2 + y^2 = 1 \quad (5)$$

From (3), we have $\lambda = 2$, using it in (1) and (2), we get

$$0 = 4 + 2\mu x \quad \& \quad 4 = -2 + 2\mu y \quad \implies \boxed{x = -\frac{2}{\mu}}, \boxed{y = \frac{3}{\mu}}$$

Using the constraint $x^2 + y^2 = 1$

Substituting $x = -\frac{2}{\mu}$ and $y = \frac{3}{\mu}$ into (5): $\left(-\frac{2}{\mu}\right)^2 + \left(\frac{3}{\mu}\right)^2 = 1$

$$\frac{4}{\mu^2} + \frac{9}{\mu^2} = 1 \quad \Rightarrow \quad \mu \pm \sqrt{13}$$

$\mu = \sqrt{13}$ for $x = -\frac{2}{\mu}$ and $y = \frac{3}{\mu}$

$$x = -\frac{2}{\sqrt{13}}, \quad y = \frac{3}{\sqrt{13}}$$

Substituting into equation (4):

$$2x - y - z = 2$$

$$-\frac{4}{\sqrt{13}} - \frac{3}{\sqrt{13}} - z = 2 \quad \boxed{z = -2 - \frac{7}{\sqrt{13}}}$$

$\mu = -\sqrt{13}$ for $x = -\frac{2}{\mu}$ and $y = \frac{3}{\mu}$

$$x = \frac{2}{\sqrt{13}}, \quad y = -\frac{3}{\sqrt{13}}$$

Substituting into equation (4):

$$2x - y - z = 2$$

$$\frac{4}{\sqrt{13}} + \frac{3}{\sqrt{13}} - z = 2 \quad \boxed{z = -2 + \frac{7}{\sqrt{13}}}$$

Evaluating the Function f

We now evaluate the function f at both solutions:

For $x = -\frac{2}{\sqrt{13}}, y = \frac{3}{\sqrt{13}}$ and $z = -2 - \frac{7}{\sqrt{13}}$

$$f\left(-\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}, -2 - \frac{7}{\sqrt{13}}\right) = 4 + 26\sqrt{13} = 11.2111$$

For $x = \frac{2}{\sqrt{13}}, y = -\frac{3}{\sqrt{13}}$ and $z = -2 + \frac{7}{\sqrt{13}}$

$$f\left(\frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}, -2 + \frac{7}{\sqrt{13}}\right) = 4 - 26\sqrt{13} = -3.2111$$

Conclusion

We have the following solutions:

- Maximum at $\left(-\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}, -2 - \frac{7}{\sqrt{13}}\right)$
- Minimum at $\left(\frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}, -2 + \frac{7}{\sqrt{13}}\right)$

Optimization Problem Using Lagrange Multipliers

We are given the objective function:

$$f(x, y, z) = 3x^2 + y$$

with the constraints:

$$4x - 3y = 9 \quad \text{and} \quad x^2 + z^2 = 9$$

Using Lagrange multipliers, we solve the system:

$$6x = 4\lambda + 2x\mu$$

$$1 = -3\lambda$$

$$0 = 2z\mu$$

and the constraints:

$$4x - 3y = 9$$

$$x^2 + z^2 = 9$$

Step 1: Finding the value of λ

From the second equation, we can immediately solve for λ :

$$\lambda = -\frac{1}{3}$$

Step 2: Exploring $z = 0$

From the third equation, we have two possibilities: $z = 0$ or $\mu = 0$. Let's first consider $z = 0$.

Substituting into the second constraint:

$$x^2 = 9 \quad \Rightarrow \quad x = \pm 3$$

Now substitute into the first constraint to find y :

$$x = -3 : \quad -12 - 3y = 9 \quad \Rightarrow \quad y = -7$$

$$x = 3 : \quad 12 - 3y = 9 \quad \Rightarrow \quad y = 1$$

This gives two possible points:

$$(-3, -7, 0) \quad \text{and} \quad (3, 1, 0)$$

Step 3: Exploring $\mu = 0$

Now consider the case when $\mu = 0$. Substituting into the first equation:

$$6x = -\frac{4}{3} \Rightarrow x = -\frac{2}{9}$$

Substituting into the first constraint:

$$4\left(-\frac{2}{9}\right) - 3y = 9 \Rightarrow y = -\frac{89}{27}$$

Now substitute into the second constraint:

$$\left(-\frac{2}{9}\right)^2 + z^2 = 9 \Rightarrow z = \pm \frac{5\sqrt{29}}{9}$$

This gives two additional points:

$$\left(-\frac{2}{9}, -\frac{89}{27}, -\frac{5\sqrt{29}}{9}\right) \quad \text{and} \quad \left(-\frac{2}{9}, -\frac{89}{27}, \frac{5\sqrt{29}}{9}\right)$$

Step 4: Evaluating the Function at Each Point

We now evaluate the function $f(x, y, z) = 3x^2 + y$ at each of the four points:

$$f(-3, -7, 0) = 20$$

$$f(3, 1, 0) = 28$$

$$f\left(-\frac{2}{9}, -\frac{89}{27}, -\frac{5\sqrt{29}}{9}\right) = -\frac{85}{27}$$

$$f\left(-\frac{2}{9}, -\frac{89}{27}, \frac{5\sqrt{29}}{9}\right) = -\frac{85}{27}$$

Step 5: Determining the Absolute Extrema

The absolute maximum value is 28, which occurs at the point:

$$(3, 1, 0)$$

The absolute minimum value is $-\frac{85}{27}$, which occurs at the points:

$$\left(-\frac{2}{9}, -\frac{89}{27}, -\frac{5\sqrt{29}}{9}\right) \quad \text{and} \quad \left(-\frac{2}{9}, -\frac{89}{27}, \frac{5\sqrt{29}}{9}\right)$$